

N+1 Redundancy: Grid A / Grid B Cross-Backup

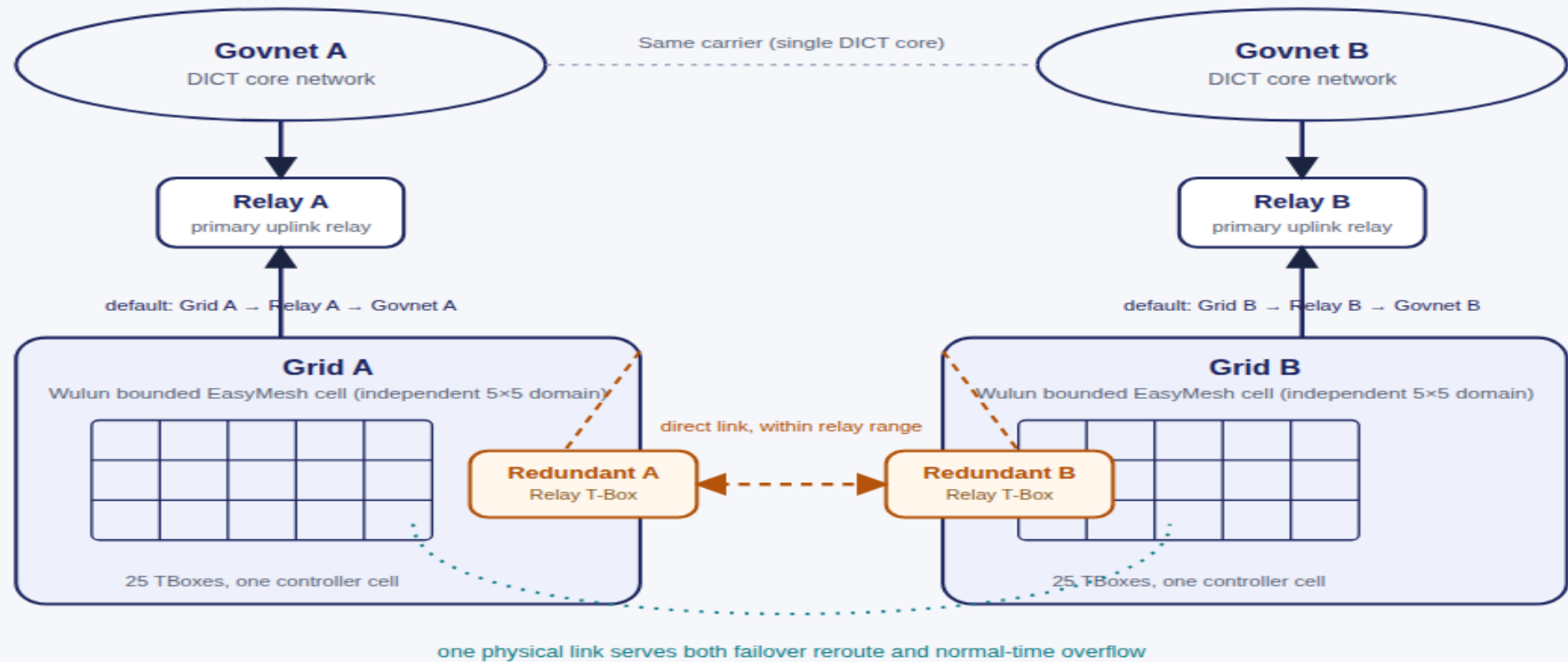
Mapped to Wulun's GVC Circuit Intent

A cross-grid backup relay link that does double duty as failover protection and normal-time overflow balancing — without requiring Grid A and Grid B to share one EasyMesh controller domain.

Design Summary

- 1 One cross-grid relay link.** Redundant A and Redundant B are dedicated Relay T-Boxes, linked directly within relay range — independent of each grid's primary Govnet uplink.
- 2 Two jobs, one link.** The same physical link serves as a failover route on Govnet outage, and as an overflow route for normal-time load spikes.
- 3 No EasyMesh merge required.** Grid A and Grid B stay fully independent bounded EasyMesh cells; the cross path exists only at Layer 3 (Babel / GVC).
- 4 Modeled as two PROTECTED circuits.** One GVC CircuitIntent per grid uplink, each with a primary path and a diversity-tagged backup path.
- 5 One open question.** Failover and load-balancing are not the same GVC mechanism — admission handling needs an explicit decision.

One Cross-Grid Link Doubles as Failover and Overflow Path



Default paths (solid): Grid A reaches Govnet A via Relay A; Grid B reaches Govnet B via Relay B. **Cross path (amber, dashed):** Redundant A (attached to Grid A) and Redundant B (attached to Grid B) are linked directly, within relay range of each other. This single link is used both as a failover route when Govnet A or B goes down, and as an overflow route in normal operation, sending excess traffic to whichever grid currently has spare uplink capacity. Grid A and Grid B remain fully independent EasyMesh controller cells; this cross path exists only at Layer 3 (Babel/GVC).

Two PROTECTED CircuitIntents Sharing One Backup Link

CircuitIntent — Grid A uplink

```
circuit_uuid: gvc-grid-a-uplink
ingress_node: Grid A egress T-Box
egress_node: Govnet A Gateway
mode: PROTECTED

primary_path:
  [Grid A → Relay A → Govnet A]

backup_path:
  [Grid A → Redundant A → Redundant B
   → Grid B → Relay B → Govnet B]

diversity: link | shared_risk
fallback: policy (BABEL | DROP)
class: split into REALTIME/PREMIUM/BEST_EFFORT by policy
```

mirror
.....

CircuitIntent — Grid B uplink

```
circuit_uuid: gvc-grid-b-uplink
ingress_node: Grid B egress T-Box
egress_node: Govnet B Gateway
mode: PROTECTED

primary_path:
  [Grid B → Relay B → Govnet B]

backup_path:
  [Grid B → Redundant B → Redundant A
   → Grid A → Relay A → Govnet A]

diversity: link | shared_risk
fallback: policy (BABEL | DROP)
class: split into REALTIME/PREMIUM/BEST_EFFORT by policy
```

Both circuits' backup_path share the same physical Redundant A ↔ Redundant B link

→ the admission ledger must treat this link as one shared resource, so the two CircuitIntents don't double-book each other's backup bandwidth

Rev.7's PROTECTED mode is defined so that backup_path activates only when the primary path fails.

Failover and Load Balancing Are Not the Same GVC Mechanism

Rev.7, Section 8.1: “No per-packet ECMP for a TCP connection ... where multiple paths are active, hash a flow or aggregate FEC consistently.” A PROTECTED backup_path is dormant until the primary fails — continuous load balancing is a different behavior.



Separate the Circuits

- Keep the PROTECTED circuit strictly for failover (dormant backup).
- Issue a second SOFT / BEST_EFFORT circuit dedicated to normal-time overflow traffic.
- Both circuits share the same Redundant A ↔ B physical link.



Admission-Policy-Based Distribution

- No separate circuit — route new admissions to the other grid by policy.
- Trigger: either grid's capacity ledger shows no headroom.
- Fixed per flow — no mid-session path change once admitted.

Either way: the Redundant A ↔ B link's bandwidth is shared by ≥ 2 logical circuits/policies, so it must be declared as one shared resource in the admission ledger to avoid double-booking.

Questions for Wulun

- 1 Admission handling**

Which of the two approaches above (separate circuits vs. admission-policy routing) fits the GVC admission model — or is there a third pattern already assumed?
- 2 EasyMesh scope**

Confirm this design lets Grid A and Grid B each stay a single bounded EasyMesh cell, replacing the earlier two-zone / overlapping-controller proposal.
- 3 Physical validation**

Relay range between Grid A and Grid B (the Redundant A ↔ B hop) needs a link-budget check before this is treated as a given assumption.
- 4 Shared-risk audit**

Verify Relay A/Redundant A and Relay B/Redundant B don't share a tower, antenna, or power feed — required for the diversity: shared_risk tag to be valid.